Mohamed Osman

Senior Backend Engineer · NestJS · PostgreSQL · System Architecture

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PROFILE

Senior backend engineer who designed and built ABS Courier's logistics platform from scratch, replacing a purchased legacy system, then owned its architecture as Principal Architect. It runs 20M+ shipments a year for 50,000+ users: 1,000+ staff, 5,000+ business customers and the consignees tracking their shipments. Strongest in TypeScript, Node.js and NestJS, PostgreSQL performance (partitioning, read replicas, query tuning), queue-based processing with Bull and Redis caching. Led a team of five backend engineers, and still ships React, Next.js and React Native when a product needs it.

20M+	50K+	~3.7×	5
shipments a year	platform users	faster responses with Redis	backend engineers led

EXPERIENCE

ABS Courier

Cairo, Egypt · May 2024 – Oct 2026

Principal Architect Aug 2026 – Oct 2026

- Owned the architecture of the platform I designed and built, as it scaled past 20M+ shipments a year.
- Led its re-architecture into a multi-tenant SaaS for external courier companies: tenant isolation, data boundaries and API versioning.
- Set backend architecture standards across the team: decision records, review gates and zero-downtime migration practice.

Backend Team Leader Nov 2024 – Aug 2026

- Designed and built the company's logistics platform from scratch, replacing a purchased legacy system; it handles 20M+ shipments a year for 50,000+ users.
- Gathered requirements directly from operations, branch and fleet management teams and turned them into system design.
- Partitioned high-volume shipment tables by date, keeping query times flat as the dataset grew.
- Introduced PostgreSQL read replicas behind PgBouncer, routing lag-tolerant reporting reads off the primary.
- Ran zero-downtime migrations on live high-volume tables.
- Moved bulk ingest, WhatsApp notifications and tracking events onto Bull queues processed by a dedicated worker.
- Containerised the stack with Docker and introduced a Redis caching layer, improving response times ~3.7×.
- Built the public shipment-tracking and partner-facing APIs.
- Led a team of five backend engineers: review standards, onboarding and mentoring.

Full Stack Developer May 2024 – Nov 2024

- Joined while the company ran a purchased legacy system; proposed NestJS and PostgreSQL as the foundation for a purpose-built replacement and owned the architecture decisions behind it.
- Took ownership of the backend within six months and moved into the team lead role.

Digital Egypt Cubs Initiative (DECI)

Giza, Egypt · Jan 2024 – Sep 2026

Instructor

- Taught coding and robotics to young learners as part of the Digital Egypt Cubs Initiative.

NasNav

Al Rehab City, Egypt · Jun 2023 – May 2024

Software Developer

- Built web applications with React.js and cross-platform mobile apps with React Native, integrating with backend APIs.

Teqneia ICT

El Sheikh Zayed, Egypt · Jun 2022 – Jun 2023

Software Developer

- Built responsive React.js web applications for client projects.

SKILLS

Backend & architecture	Node.js (NestJS, Express), TypeScript, REST & GraphQL APIs, event-driven design, Bull queues & background workers, multi-tenant architecture
Databases	PostgreSQL (partitioning, replication, indexing, query tuning), PgBouncer, Redis (caching, queues), MySQL, MongoDB
Infrastructure	Docker, AWS (EC2, S3, RDS, Aurora), Nginx, CI/CD, OpenTelemetry, Prometheus, Loki, Git
Frontend	React.js, Next.js, React Native
Practices	Architecture decision records, zero-downtime migrations, code review, technical mentoring

EDUCATION

Bachelor of Computer Science, Faculty of Science, Cairo University
Giza, Egypt · 2018 – present

AWARDS & LANGUAGES

ECPC 2022, Egyptian Collegiate Programming Contest

Arabic (native) · English (proficient)